ENME 561/661

Lab 4



© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Second Order Systems

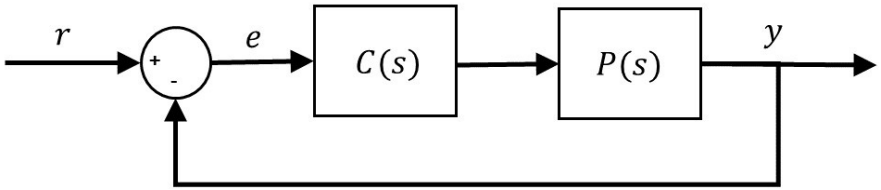
- A second-order system has a transfer function in which the order of the denominator is 2 (the largest order of s)
- General form is

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

The denominator has two poles

ω_n : natural frequency

ζ: damping ratio



© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

s-plane plot for a pair of complex poles

Complex poles with $\sigma > 0$

$$s = -\sigma \pm j\,\omega_d$$

Complex poles always come in complex conjugate pairs:

$$a(s) = (s + \sigma - j\,\omega_d)(s + \sigma + j\,\omega_d)$$
$$= (s + \sigma)^2 + \omega_d^2$$

Comparing this with the denominator of the general form of a second order system:

$$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)}$$

We find

$$\sigma = \zeta\omega_n$$
$$\omega_d = \omega_n\sqrt{1 - \zeta^2}$$

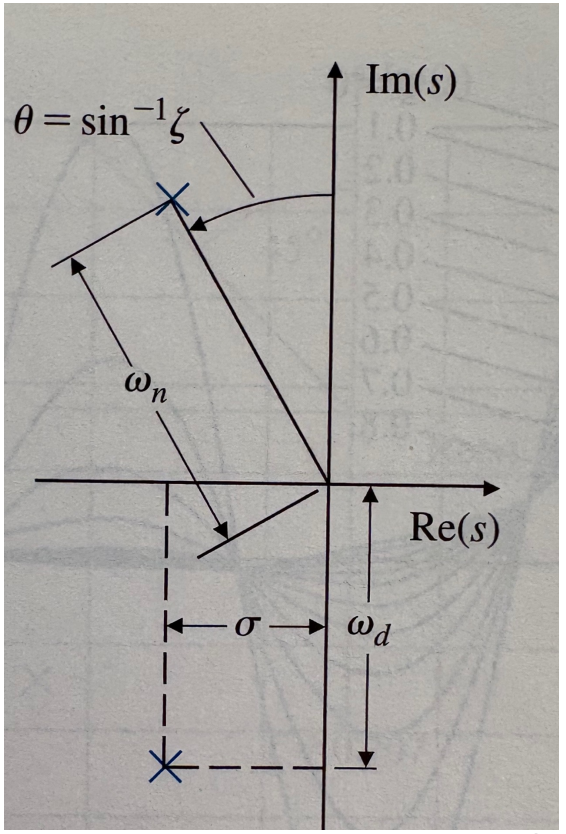


Figure source: Franklin, Powell, & Emami-Naeini, Feedback Control of Dynamic Systems, 8th ed.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Pole locations corresponding to three values of damping ratio $\zeta = \sin(\theta)$

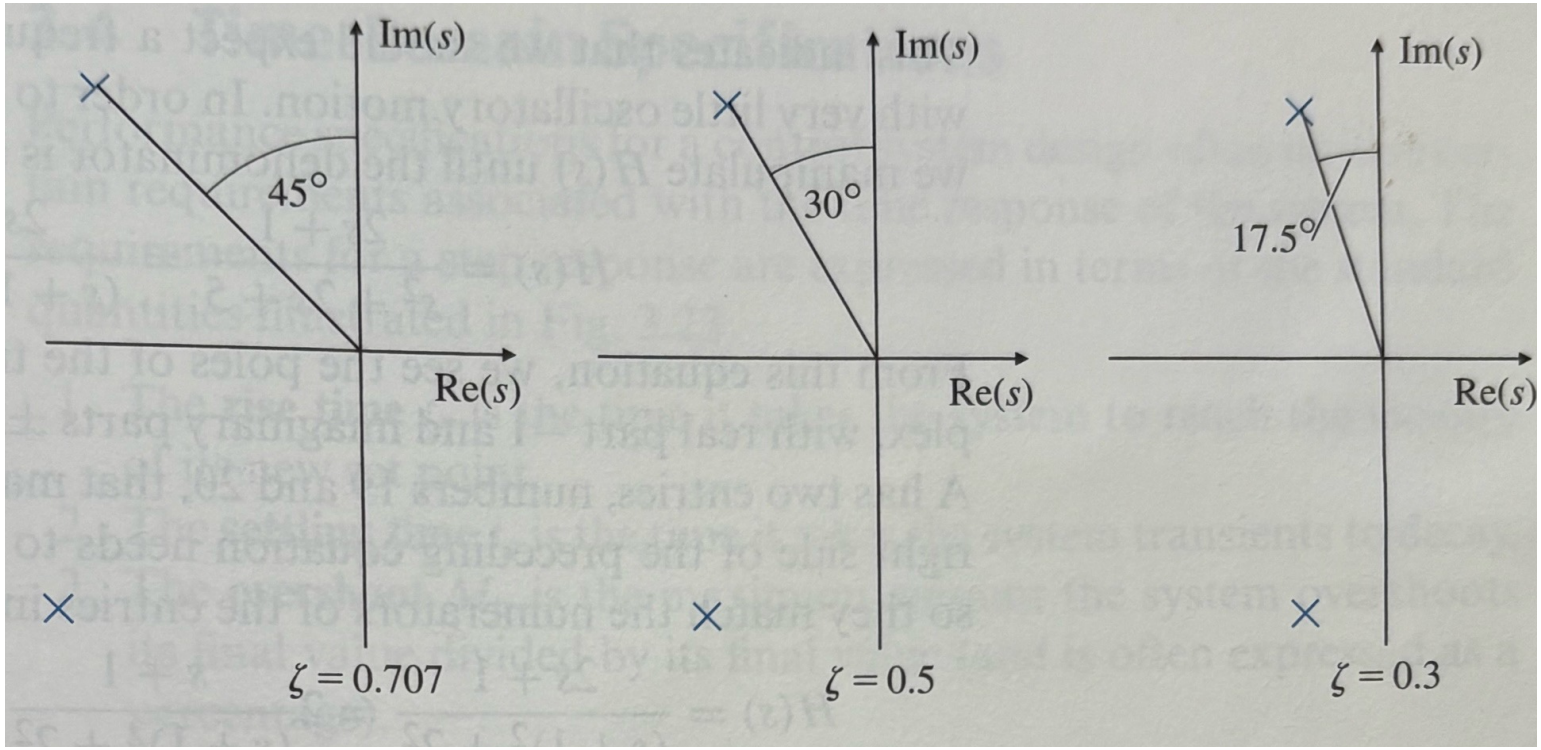


Figure source: Franklin, Powell, & Emami-Naeini, Feedback Control of Dynamic Systems, 8th ed.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Responses of a second-order system with different damping ratios

- $0 < \zeta < 1$ ➔ oscillatory motion with gradually decreasing or increasing amplitude

- $\zeta = 1$ ➔ critically damped, no oscillation

- $\zeta = 0$ ➔purely oscillatory motion

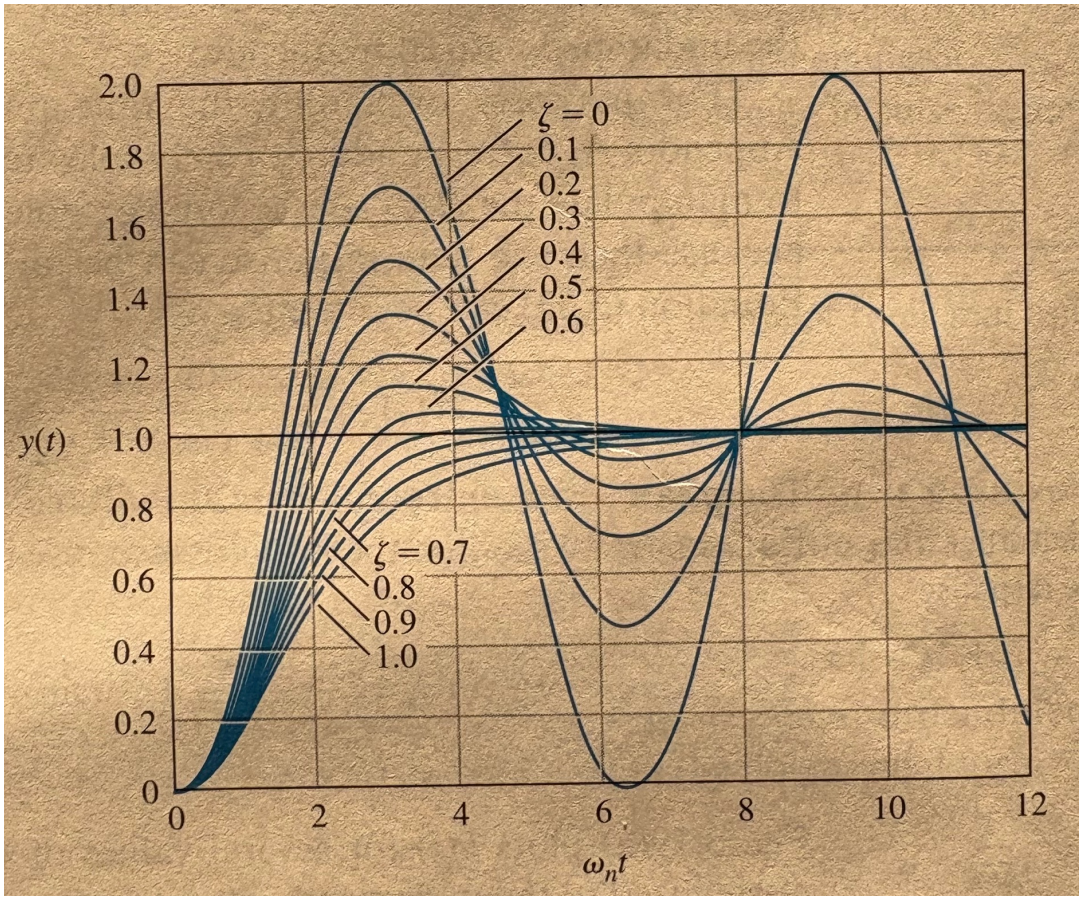


Figure source: Franklin, Powell, & Emami-Naeini, Feedback Control of Dynamic Systems, 8th ed.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Time functions associated with points in the s-plane

Larger natural frequency (larger values on the vertical axis)➔ faster oscillation

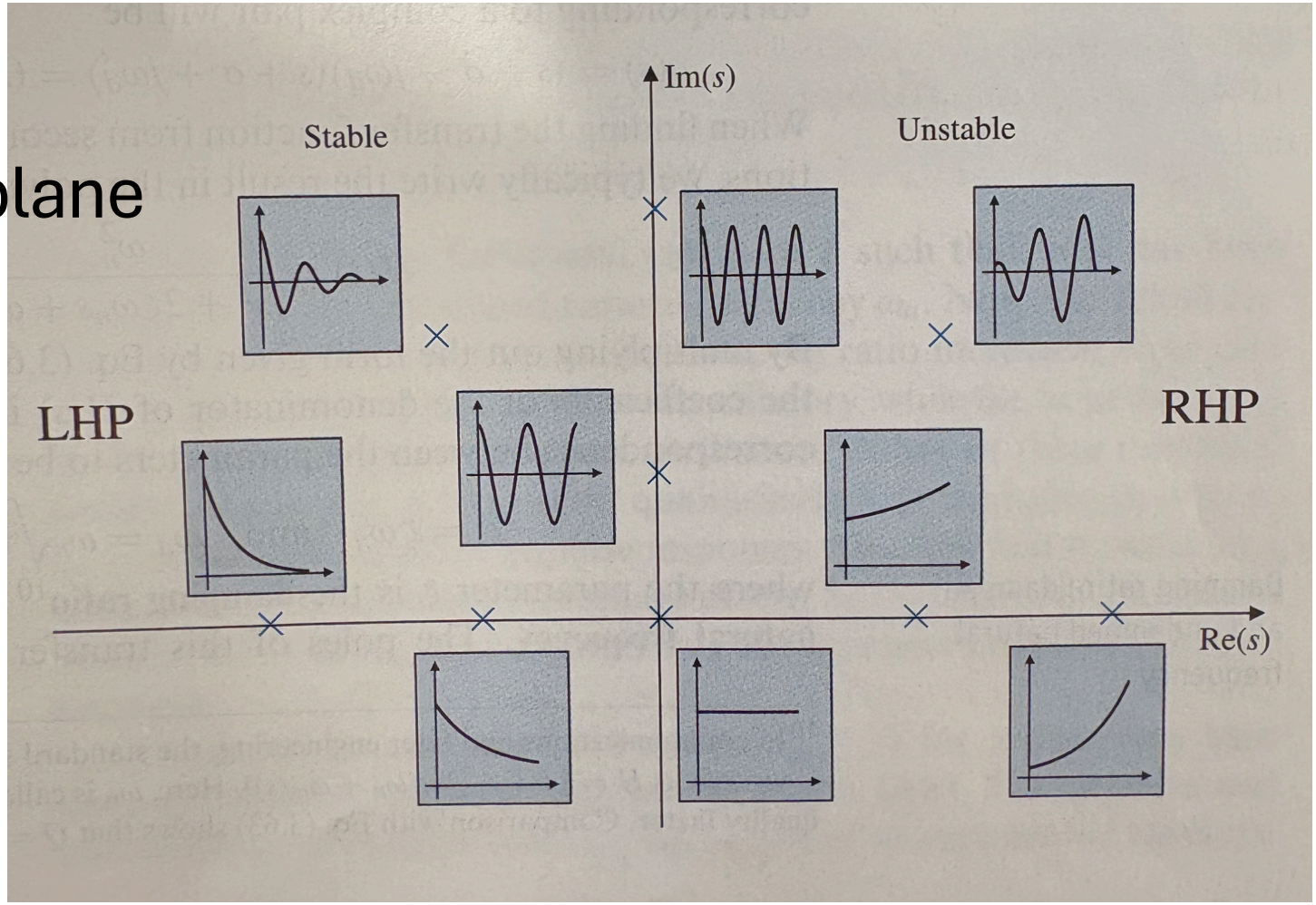


Figure source: Franklin, Powell, & Emami-Naeini, Feedback Control of Dynamic Systems, 8th ed.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

System Response

- Percent Overshoot:
  - Relative amount you are going over the target signal

$$PO = \frac{100(y_{max} - R_0)}{R_0}$$

- Peak Time:
  - Delay in the system from the reference signal

$$t_p = t_{max} - t_0$$

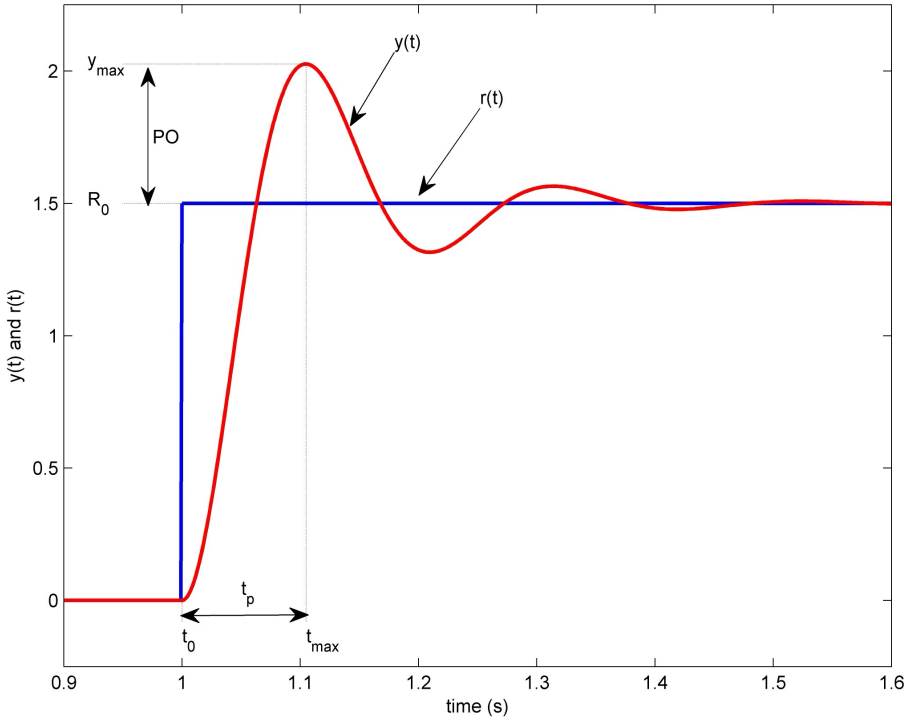


Figure source: Quanser, QUBE-Servo 2 Workbook – Student, v1.2 (2021).

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Second Order System Response

- Percent Overshoot:
  - Relative amount you are going over the target signal

$$PO = 100e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)}$$

- Peak Time:
  - Delay in the system from the reference signal

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

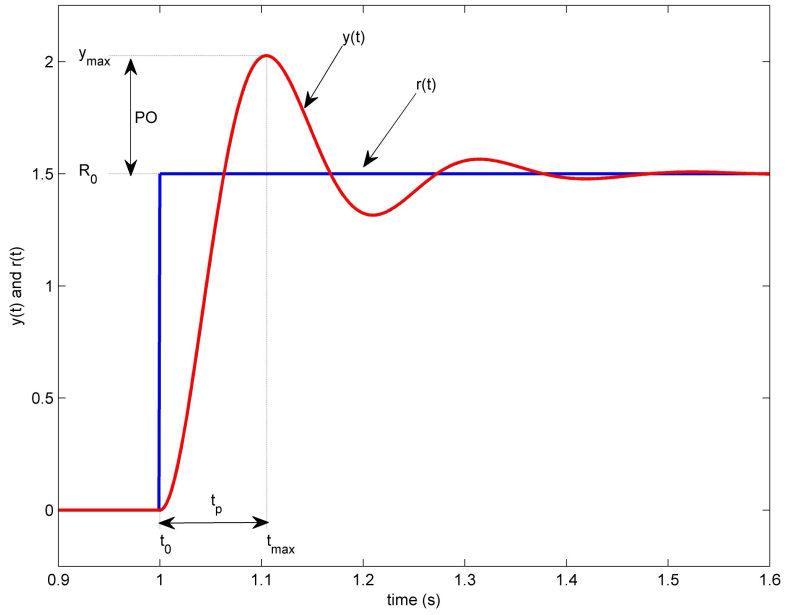


Figure source: Quanser, QUBE-Servo 2 Workbook – Student, v1.2 (2021).

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

DC Motor Model with the Unity Feedback

- Motor Model:

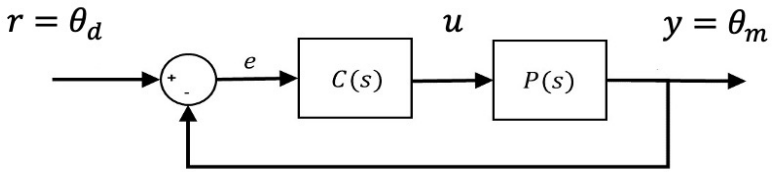
$$P(s) = \frac{\Theta_m(s)}{V_m(s)} = \frac{K}{s(\tau s+1)}$$

- K is the model steady state gain, τ is the model time constant
- Unity Feeback only, i.e. no PID controller:

$$C(s) = \frac{U(s)}{E(s)} = 1$$

- Closed Loop Transfer Function:

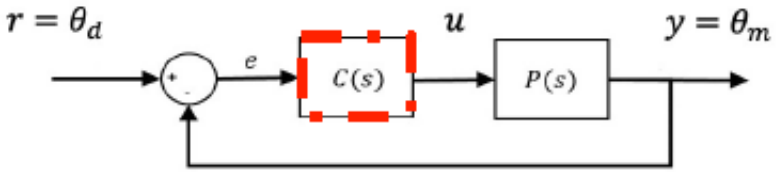
$$\frac{\Theta_m(s)}{\Theta_d(s)} = \frac{\frac{K}{\tau}}{s^2+\frac{1}{\tau}s+\frac{K}{\tau}}$$



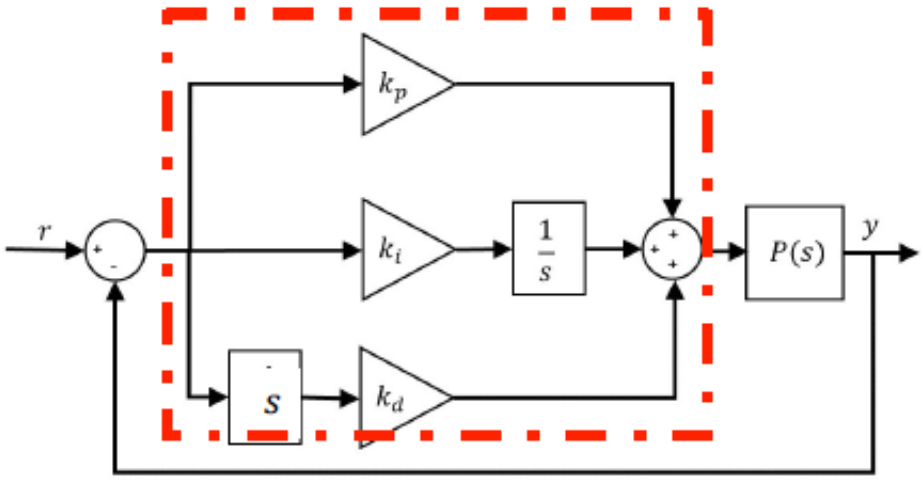
Unity feedback: the gain on the feedback signal is 1

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

PID Control



- Commonly used in industry
- Based on the error between a reference value $(r)$and the system output (y)
- Three parts
  - Proportional to account for current error
  - Integral to account for the history of the error
  - Derivative to account for predicting the future of error

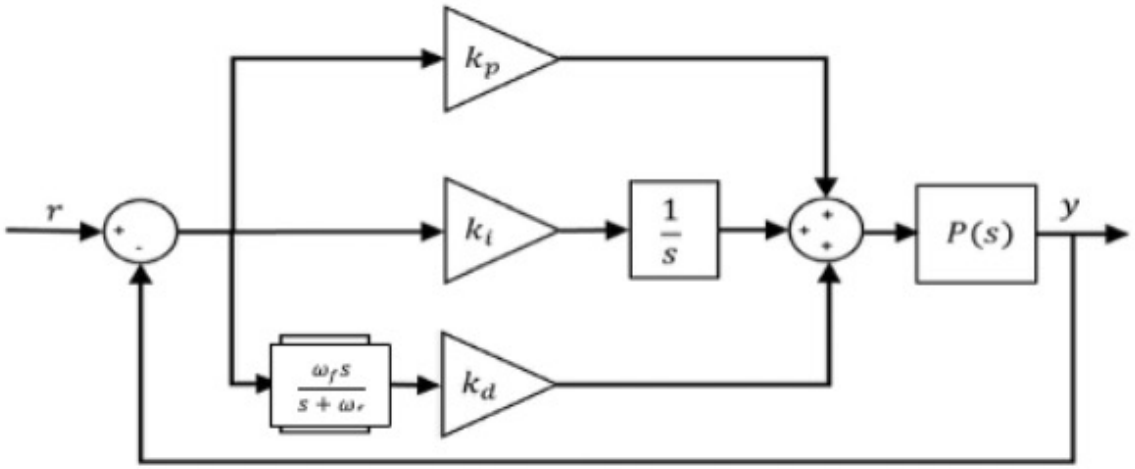


$$C(s) = \frac{U(s)}{E(s)} = k_p + \frac{k_i}{s} + k_d s$$

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

PID Control with a filter on the derivative action

First order filter: $F(s) = \frac{\omega_f}{s+\omega_f}$



© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

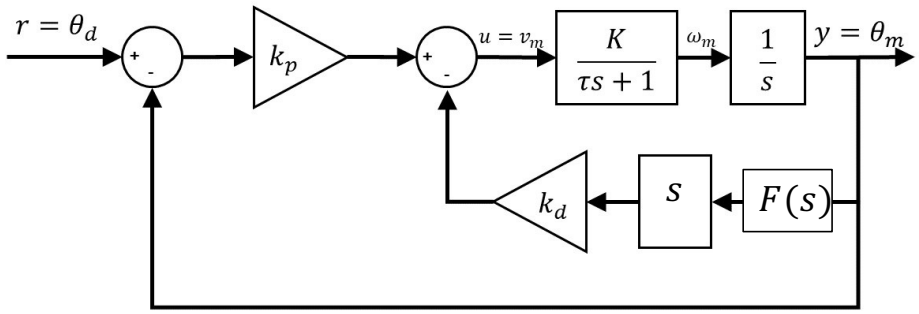
PV Control

- Modified PD control system that reduces the possibility of sudden large actuations to prevent damage to the system
- Derivative of just the position $(-y)$ and not the error($e = r - y$) is used in the feedback for this controller
- Best to use a combined filter and derivative in this case instead of a simple derivative block

$$D(s) = k_d s\, F(s) = k_d \frac{\omega_f s}{s+\omega_f}$$

Recall: First order filter:

$$F(s) = \frac{\omega_f}{s + \omega_f}$$



© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

In-Lab Exercises

6.1 Numerical Differentiation in Noisy and Rapidly Varying Signals

- MATLAB only — a Live Script, no Simulink and no QUARC blocks

- **6.2 Second Order System Step Response with the Unity Feedback**
  - Given K and τ from the data sheet, calculate ωn and ζ, then predict peak time and percent overshoot
  - Implement unity feedback on the hardware, apply a step input, and measure both experimentally
- **6.3 Second Order System Step Response with PD and PV Controllers**
  - Bench work — not graded. Only the .slx model is submitted
  - Express kp and kd as a function of ωn and ζ; see what kp, kd and the filter cut-off do to the motor
- **7 Controller Design to Specification**
  - Design kp and kd to meet a stated specification, test on the hardware, and iterate

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Section 7: Design to Specification

- **New this year, and the largest single part of the mark.**
- The specification is given: PO ≤ 2.5 %, peak time ≤ 0.15 s, and the motor must not saturate.
- Fill in Table 1 before building anything — the requirements, the constraints, and the gains they imply.
- Predict kp and kd, test on the hardware, measure what actually happens, and revise.
- Record every iteration in Table 2, including the ones that failed, with a one-line reason for each change.
- The first design will almost certainly miss. That is expected — the reasoning behind each revision is what is graded, not how fast you converge.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

What is graded, and what you submit

- 6.1 Numerical differentiation — .mlx file and a PDF export of it. MATLAB only.
- 6.2 Unity feedback — results and discussion in the main report PDF.
- **6.3 PD and PV controllers — not graded. Do the bench work; submit only the .slx.**
- 7 Controller design to specification — Tables 1 and 2, in the main report PDF.
- 7.5 Optional integral action — not graded.
- **Submit: main report PDF, .mlx, PDF export of the Live Script, and the .slx model(s).**

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Learning Outcome

1. You observe how differentiation amplifies noise and sudden changes in a signal. This analysis is important because the derivative term in a PID controller requires taking the derivative of a signal.
2. From this experiment, we learn that ω_n and ζ are key characteristics of a system's response.
3. With **unity feedback**, these parameters are fixed by the system.
4. However, by implementing **PD** or **PV controllers**, we can **design** suitable k_p and k_d values to achieve desired ω_n and ζ, thereby **shaping the system response**.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

Safety



Be very careful when increasing the voltage



Make sure to use the stall detection block properly

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only

References

- Quanser. QUBE-Servo 2 Workbook – Student, version 1.2. Quanser Inc., 2021.
- Franklin, G. F., Powell, J. D., & Emami-Naeini, A. Feedback Control of Dynamic Systems, 8th ed.

© 2026 Mahdis Bisheban | ENME 561/661 | Course use only